

Federal Flight Agency Network

Introduction

You have been assigned to design a mission-critical network for Federal Flight Agency Network, a next-generation aerospace organization developing a fully autonomous aircraft fleet. The network must support ultra-reliable, low-latency communication, integrating real-time telemetry, AI navigation, ground command stations, secure satellite links, and emergency override networks.

Each major unit is divided into sub-operational systems: the Command Center (HQ) for flight control, AI Navigation Unit for autonomous path computation, Aircraft Simulation Lab for testing, Maintenance Hangar for diagnostics and repair, and Emergency Control Node as backup. All units must communicate reliably with strict redundancy and failover.

Unit	Sub-Units	Devices
Command Center (HQ)	Control Wing	2000
AI Navigation Unit	ML Engine, Data Processing Cell	850
Aircraft Simulation Lab	Simulation Core, Testing Sandbox	1200
Maintenance Hangar	Diagnostics Bay, Repair Unit	600
Emergency Control Node	Backup Core	80

Requirements

Topology Design Services & Constraints

- Each **main unit and sub-unit** must be logically segmented using routers and/or switches.
- The **Command Center (HQ)** connects directly to Simulation Core and Emergency Node.
- AI sub-units, Simulation Sandbox, and Hangar sub-units are interconnected through a backbone switch network.
- Emergency Node must **NOT** directly connect to any AI sub-unit.
- Hangar units can access HQ only through the Simulation Lab.
- At least two redundant paths must exist between the AI Unit and HQ using different sub-units.
- Only HQ hosts centralized DNS and Web services accessible via www.federalfft.control, displaying the message “*Autonomous Flight Control System Active.*”
- Email communication must be configured between mail.hq.federal and mail.ai.federal using SMTP and POP3.
- Simulation Core hosts a DHCP server, while AI and Hangar routers provide DHCP services locally, and Emergency Node uses only static addressing.
- Each sub-unit must include at least one printer and two PCs.
- Default routes are NOT allowed, and routing must ensure strict isolation and controlled redundancy.

Addressing

Base network must be generated dynamically based on group size:

Group Size	Combination Rule
2 members	Last 2 digits of the 2nd member's student ID
3 members	Middle 2 digits of the 3rd member's student ID
4 members	Sum of the last 3 digits of the 4th member's ID

The extracted digits are split into two parts to form the first two octets, while the remaining octets are 0 with a /16 prefix. For example, if the group has 3 members and the 3rd member's ID is 20211121, the middle 2 digits (11) are taken and split into 1 and 1, resulting in the network 1.1.0.0/16. Apply **VLSM subnetting** considering all units and sub-units. HQ and Emergency Node use static IPs, Simulation Lab uses DHCP via a server, and AI Unit and Hangar use DHCP via routers.

DHCP Configuration

- The Simulation Core router must forward requests to its DHCP server.
- AI and Hangar routers must allocate IPs dynamically per sub-unit.
- Emergency Node must not use DHCP under any condition.

Routing Design

- HQ to Emergency Node must use next-hop static routing.
- Emergency Node to HQ must use exit-interface static routing.
- Simulation Lab routers must use static routing toward HQ and dynamic RIPv2 routing toward AI and Hangar sub-units.
- Floating static route from AI Unit to HQ via Simulation Sandbox must use an Administrative Distance (AD) calculated as: (sum of digits of the first member's student ID) $\times$ (number of group members)
- Hangar must not learn HQ routes directly from the AI Unit.
- Emergency Node must not participate in dynamic routing.
- Routing must prevent loops and ensure redundancy without default routes.

★ You **are allowed to make** any valid and necessary assumptions while designing the network infrastructure.

Deliverables

1. Complete implementation in Cisco Packet Tracer
2. Full end-to-end connectivity validation
3. Submission must include:
 - a. Packet Tracer (.pkt/.pka) file
 - b. Topology diagram with labeled subnets and sub-units
 - c. VLSM tree
 - d. IP addressing table
 - e. Full configuration commands for routers and servers